



Abstract; to make sure you're in the right room

Have you tried to find the most effective way to ingest and process your data?
Were you just as confused as I was after reading all the marketing materials?
Would you like to gain more insight into maximising your capacity?

These are questions I've had. And ones I regularly try to help customers with.

In this session, I'll help you learn the differences between Lakehouse, SQLDB, and Warehouse performance, processing speed and cost. I'll also demonstrate how the Realtime stack can serve as a viable alternative for ingestion and processing flat files in terms of speed, scalability and ease of use.

After this session, you'll walk away with real-life insights from Fabric and knowledge of which buttons not to press.

If you've attended the previous instalment of this session, you'll know I won't do my testing with AdventureWorks. It will be a lot of (messy) data to give Fabric a run for its capacity units!





Loadtesting Fabric, part II

The sequel

Reitse Eskens

When stuff goes off the rails



What happened previously





The main findings

Lakehouses
were very
good

Warehouses
were very slow
and expensive

Dataflows
Gen2 only
with FastCopy



F2 or F64

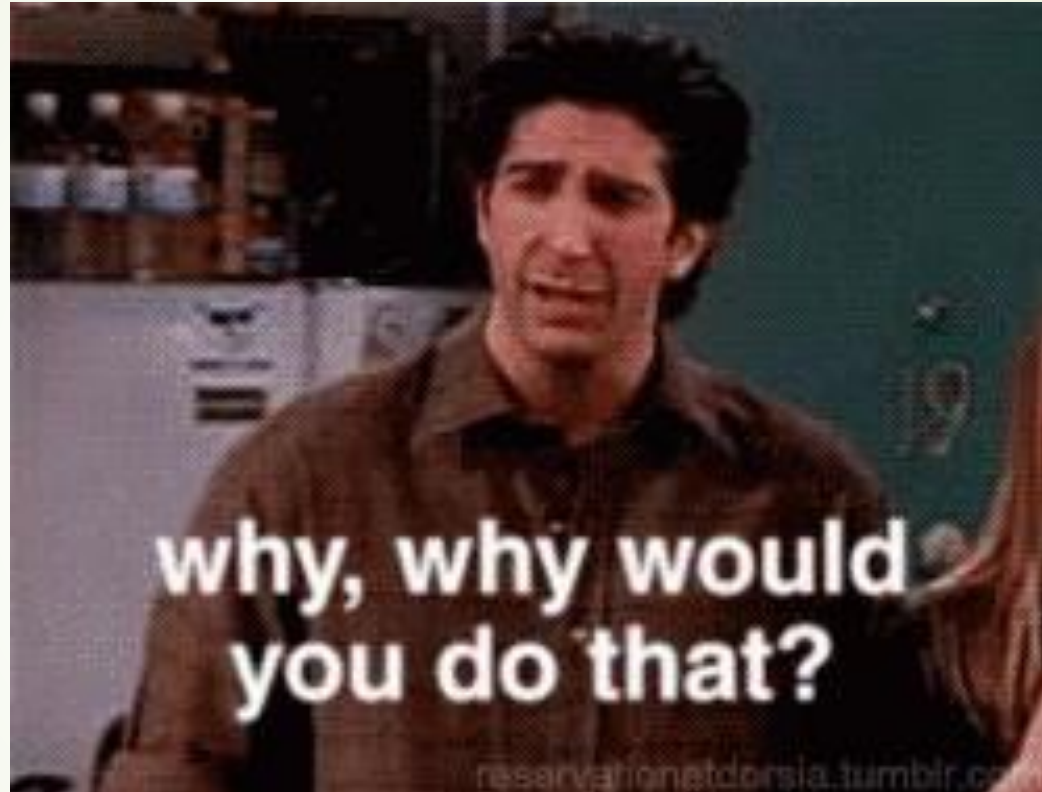
- With a lot of data, big differences in processing
 - Mostly time
 - Sometimes capacity exhaustion



Let's do it again!

Break... I mean loadtesting Fabric, again

Why



Because reasons



WHEN WILL IT BREAK



HOW WILL IT BREAK



WHAT'S THE MOST
EXCITEMENT FOR YOUR
EURO

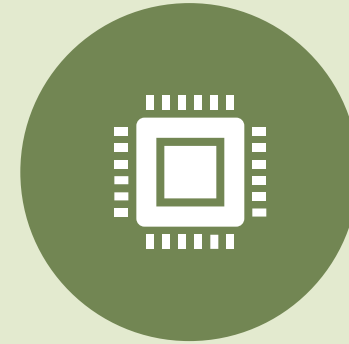
No happy flows



(AZURE) SQL DB IS QUITE
EASY

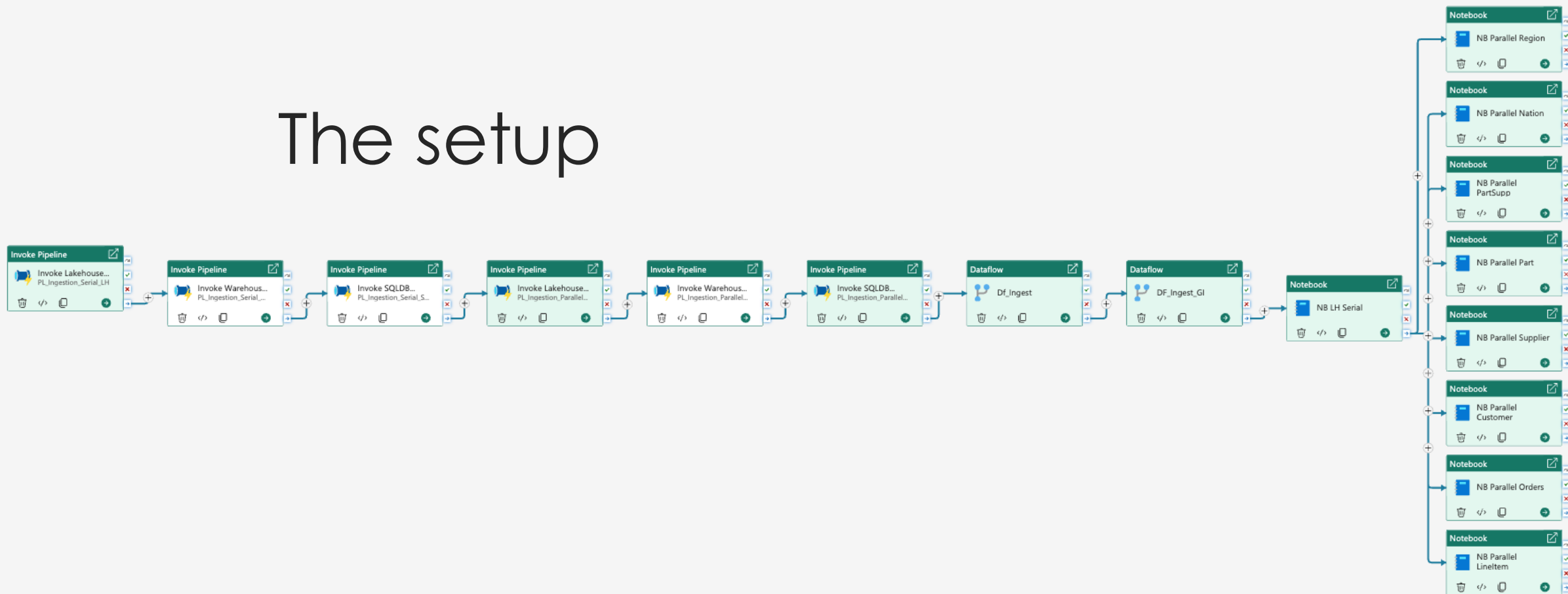


ADVENTUREWORKS
DATABASE IS TOO SMALL



FREE API'S ARE USUALLY
VERY SIMPLE

The setup



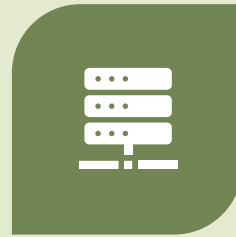
The setup



ONE MASTER PIPELINE



PARALLEL AND SERIAL
DATA INGESTION



LAKEHOUSE,
WAREHOUSE, FABRIC
SQL DB AND
EVENTHOUSE



COPY ACTIVITY,
NOTEBOOKS,
DATAFLows GEN2,
EVENTSTREAM

The data



500ISH GIGABYTE OF
TPC-H DATA



SMALL FILES (FEW KB)
TO LARGE ONES (75 GB)



ALL CSV, AS GOOD AS
IT GETS

Metrics



FABRIC SQL DB
DASHBOARD



FABRIC
CAPACITY
METRICS APP



FABRIC
MONITORING



COST
CALCULATION

Metrics



FABRIC SQL DB
DASHBOARD



FABRIC UNIFIED
ADMIN
MONITORING



FABRIC
MONITORING



COST
CALCULATION

FUAM is your best friend



Open Source and free to use



A lot of details



Worth your investment in time (and CU(s))



<https://github.com/microsoft/fabric-toolbox/tree/main/monitoring/fabric-unified-admin-monitoring>

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/in/reitseeskens



<https://sqlreitse.com>



@2meterDBA





Storage options



Lakehouse

The Lakehouse (not the one made by Lego)



SPARK



HIGH CODE



CONFIGURABLE



Warehouse

Warehouse (you know where to find it)

SQL 'shell'
around spark

Low code and
high code
approaches

Full T-SQL (still)
not supported



Eventhouse



Eventhouse (not just a party place)

KQL

Can
synchronise
to OneLake

Blazing fast,
at a cost



Fabric SQL Database



Fabric SQL (not your average SQL)

Azure SQL
made
available in
Fabric

0 configuration

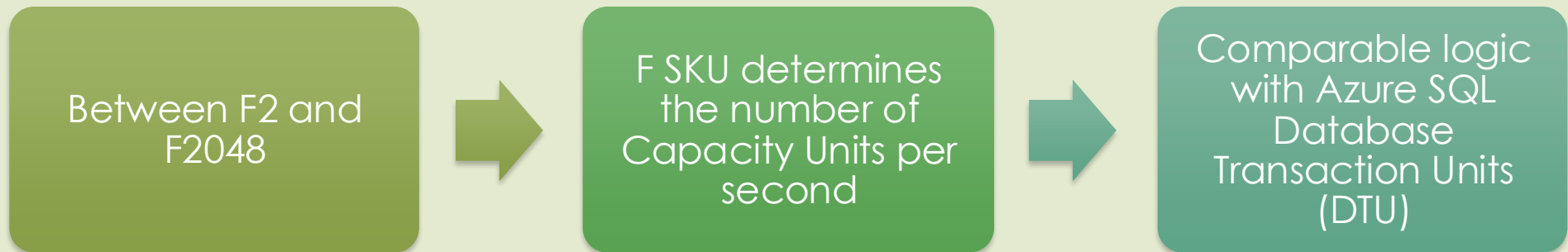
It's a different
story



Capacity and Cost



Capacities



Fabric Capacity and cost

F2 – F2048

Higher = more
power

Every step = X2

Reservations
can be
applied on
yearly basis

Fabric Capacity and cost



F2 = 60 CU(s) per 30 Seconds



F2 = $60 * 2 * 60 * 24 = 172.800$ per day



F2 costs 272, 97 euro each month, 9 euro each day



1 CU = $9 / 172.800 = 0.00005$ euro



60 CU(s) = 0,003 euro

Fabric Capacity and cost



F64 = 1920 CU(s) per 30 Seconds



F64 = $1920 * 2 * 60 * 24 = 5.529.600$ per day



F64 costs 8.734,93 euro each month, 291 euro each day



1 CU = $291 / 5.529.600 = 0.00005$ euro



1920 CU(s) = 0,101 euro

More power?





Ingestion methods



Ingestion
(are you
hungry
too?)

Pipeline

Dataflow

Notebook

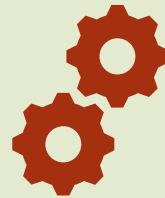
Eventstream



Pipeline



Think Azure Data
Factory



Orchestration, simple
transformations and
data ingestion

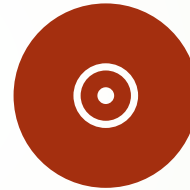


Use it wisely

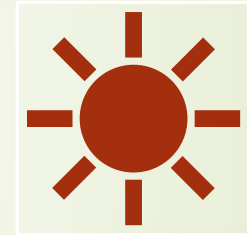
Dataflow Gen2



Low code solution



The new CI/CD is faster
and more efficient



Use the cool options

Dataflow Gen2

Options

Global

General

Data load

Diagnostics

Dataflow

Data load

Regional settings

Privacy

Scale

Parameters

Fast Copy

☒ Allow use of fast copy connectors

Partitioned compute (Preview)

☒ Allow use of partitioned compute

[Learn more](#)

Query Evaluation (Preview)

☒ Allow use of the modern query evaluation engine

[Learn more](#)

Concurrency ⓘ

☐ Limit number of concurrent evaluations

4



ool options

Notebook



HIGH CODE

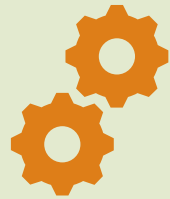


DEFAULT PYSPARK, OTHER
LANGUAGES AVAILABLE



DO WHATEVER YOU
WANT

Eventstream



Managed process to
ingest into Realtime



Configuration depends
on your source data

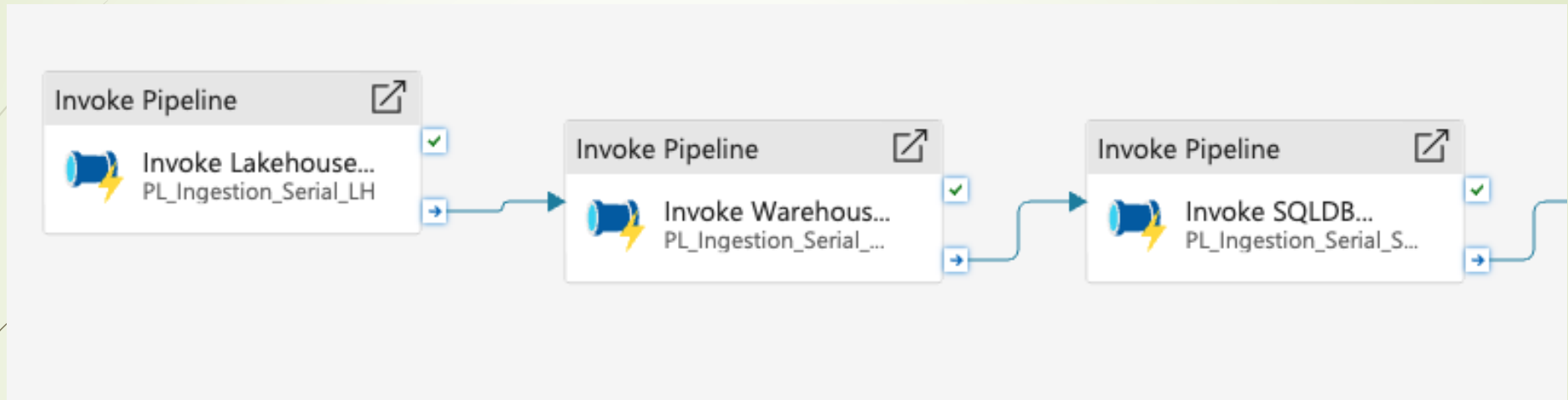


Use functions inside the
Eventhouse to model
and prepare

Let's go!

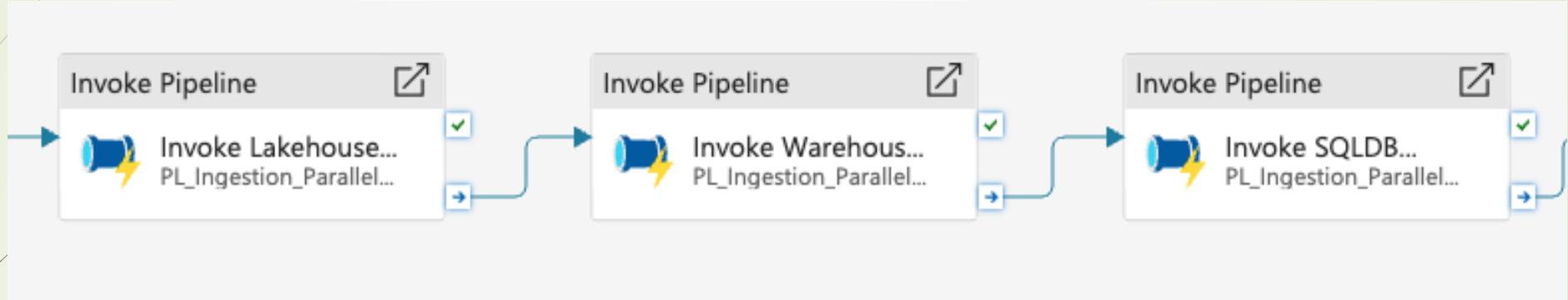


Pipeline to destination (serial)



Destination	Time	Capacity Units	Cost in Euro's
Lakehouse	1h 42m	499.320	24,96
Warehouse	2h 3m	563.400	28,17
Fabric SQL DB	4h 54m	1.376.100	68,80

Pipeline to destination (parallel)



Destination	Time	Capacity Units	Cost in Euros
Lakehouse	5h 42m	355.680	17,78
Warehouse	1h 23m	116.100 (ERROR)	5,80
Fabric SQL DB	1h 18m	867.780(ERROR)	43,39

Pipeline to destination (parallel)

Error details

 Copy Order

Details

ErrorCode=UserErrorWithLargeConcurrentRuns,'Type=Microsoft.DataTransfer.Common.Shared.HybridDeliveryException,Message=There are substantial concurrent copy activity executions which is causing failures due to throttling under subscription 81b5250a-6a84-4305-97d5-f7f98d1d911e, region we and limitation 400. Please reduce the concurrent executions. For limits, refer <https://aka.ms/adflimits..> ErrorCode: UserErrorWithLargeConcurrentRuns.,Source=Microsoft.DataTransfer.TransferServiceClient,'Type=System.Net.Http.HttpRequestException,Message=Response status code does not indicate success: 400 (Bad Request).,Source=System.Net.Http,'

Are you satisfied with your experience?  

Capacity Units

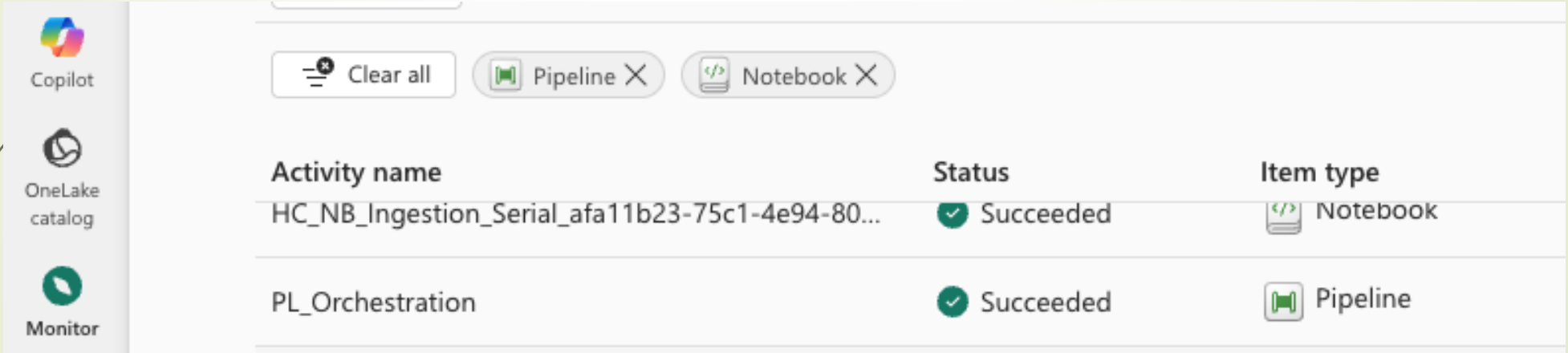
355.680

116.100 (ERROR)

867.780(ERROR)

Why the 400?

➡ Let's go into the monitoring

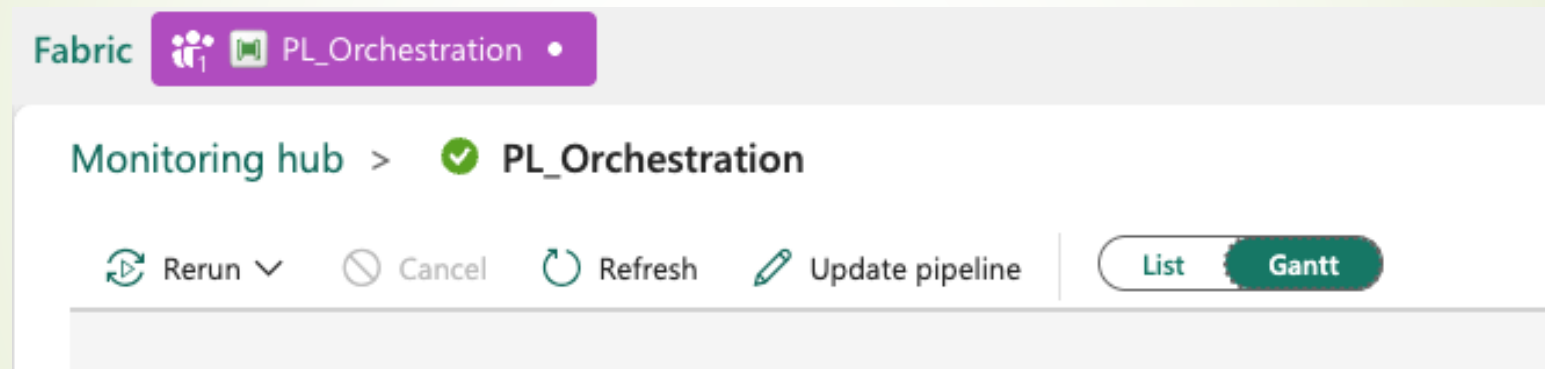


The screenshot shows the Azure Monitor interface. On the left is a sidebar with icons for Copilot, OneLake catalog, and Monitor. The main area displays a table of activities. At the top of the table are three filters: 'Clear all', 'Pipeline', and 'Notebook'. The table has three columns: 'Activity name', 'Status', and 'Item type'. Two activities are listed: 'HC_NB_Ingestion_Serial_afa11b23-75c1-4e94-80...' and 'PL_Orchestration', both with a 'Succeeded' status.

Activity name	Status	Item type
HC_NB_Ingestion_Serial_afa11b23-75c1-4e94-80...	✓ Succeeded	Notebook
PL_Orchestration	✓ Succeeded	Pipeline

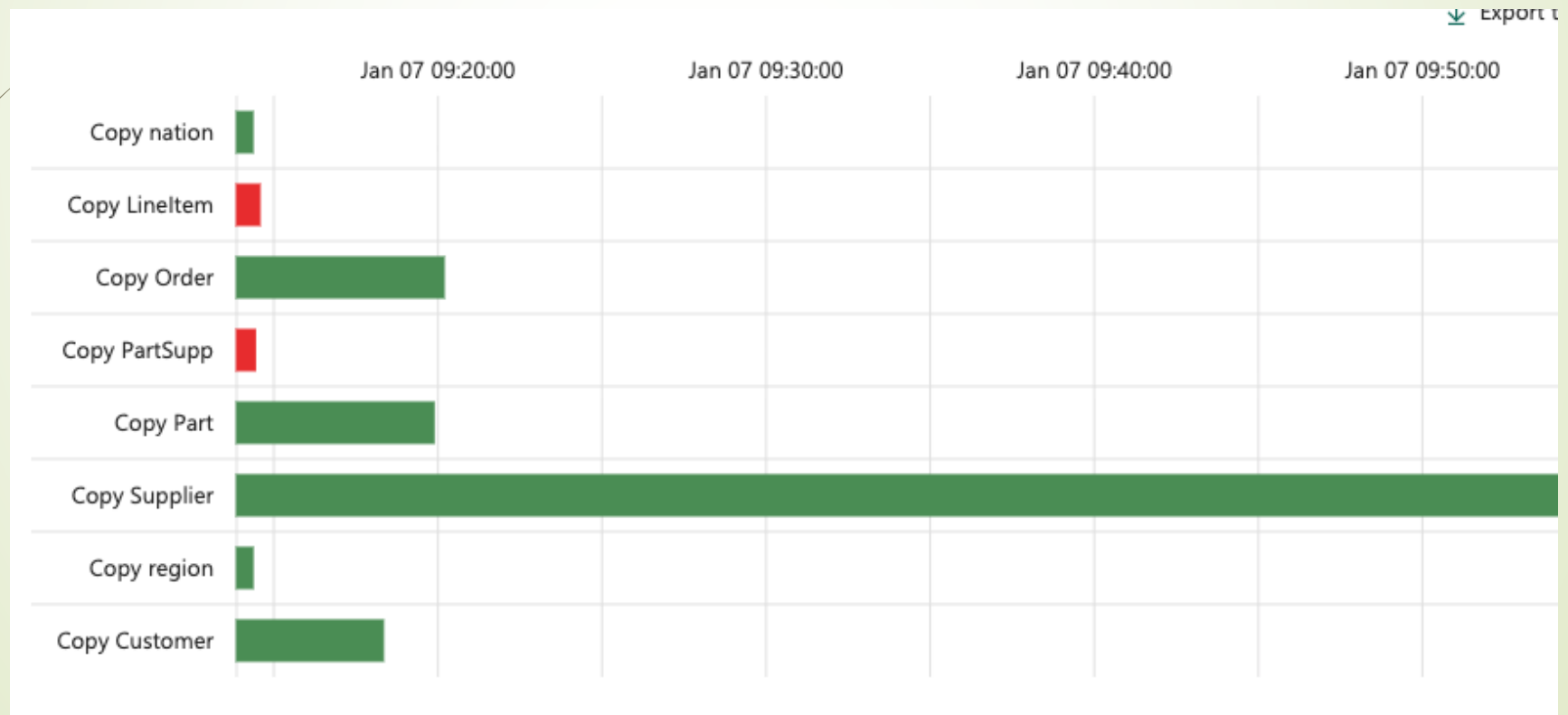
Why the 400?

➡ Let's go into the monitoring



Why the 400?

➡ Let's go into the monitoring



Why the 400?

- Let's go into the monitoring

```
"rowsRead": 5000000,  
"rowsCopied": 5000000,  
"copyDuration": 3977,  
"throughput": 182.503,  
"errors": [],  
"usedDataIntegrationUnits": 4,  
"usedParallelCopies": 2,  
"executionDetails": [  
  {
```

What 400?

Table name	Used Data Integration Units	Running total
Nation	4	4
Order	256	260
Part	44	304
Supplier	4	308
Region	4	312
Customer	36	348
LineItem	256	604  Failed 
PartSupp	228	832  Failed 

What do the docs say?

- <https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/azure-subscription-service-limits#azure-data-factory-limits>

Concurrent Data Integration Units¹ consumption per subscription per [Azure Integration Runtime region](#)

Region group 1 ² : 6,000	Region group 1 ² : 6,000
Region group 2 ² : 3,000	Region group 2 ² : 3,000
Region group 3 ² : 1,500	Region group 3 ² : 1,500



Dataflows to Lakehouse

- Error upon error
- Too many files, too much data

Dataflow Gen2	Dataflow Gen2 CI/CD
1.214.279 CU(s) (60,71 Euro)	64.469 CU(s) (3,22 Euro)
On average 4h runtime	2h runtime before error

Dataflows to Lakehouse

Error details



DF_Ingest_Gl



Error code 20302 

Failure type User configuration issue

Details Dataflow refresh job failed with status: Failed. Failure reason: { requestId: 4a6aa330-6468-4194-8bdf-a5bac2c0d720, errorCode: EntityUserFailure, message: Something went wrong, please try again later. If the error persists, please contact support. }



Nothing to see here.
Please disperse.



Notebooks and environments



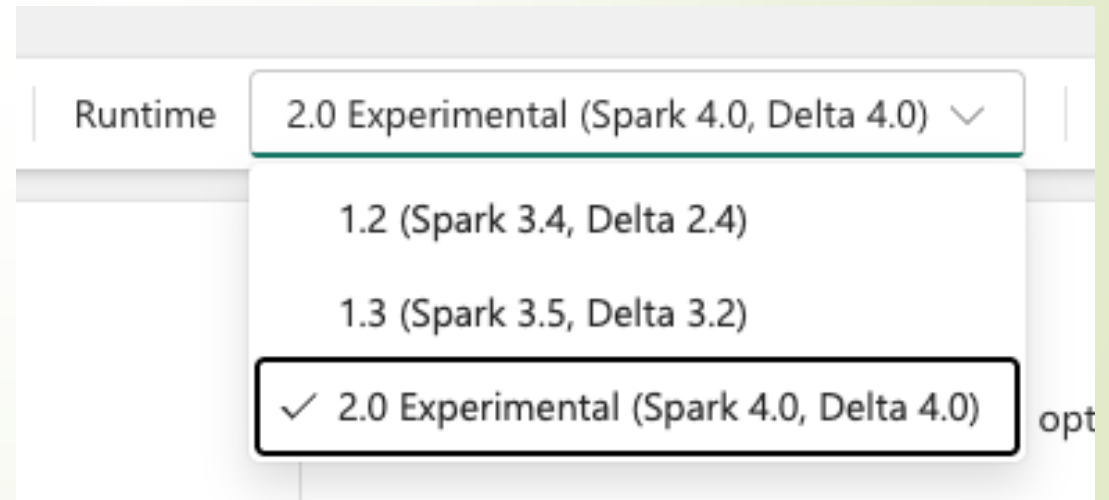
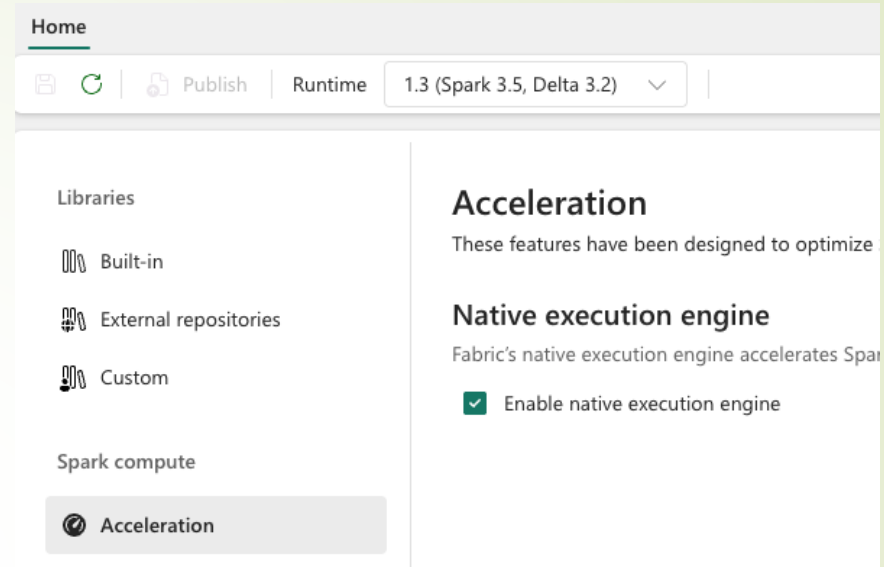
Notebook environments

Default (when you do nothing)

Native Execution Engine
(create your own environment)

Fabric Runtime 2.0 (in preview,
create your own environment)

Native Execution Engine and Fabric 2.0





Notebook to Lakehouse

Serial Time	Serial CU(s)	Parallel Time	Parallel CU(s)
24 minutes	107.627	26 minutes	??

Notebook to Lakehouse

Item Name	Item State	FUAM Deleted State	Avg CU (s) 14d	Avg CU (s) 30d	U(s)
NB_Ingestion_Nation	Active	False	15,177	15,177	
NB_Ingestion_PartSupp	Active	False	1,263	1,263	
NB_Ingestion_Customer	Active	False	48,093	48,093	
NB_Ingestion_Supplier	Active	False	40,477	40,477	
NB_Ingestion_Serial	Active	False	107,627	107,627	
NB_Ingestion_Region	Active	False	34,620	34,620	
NB_Ingestion_Part	Active	False	4,186	4,186	
NB_Ingestion_Orders	Active	False	29,711	29,711	

Notebook to Lakehouse

Serial

24 m

HC_NB_Ingestion_Nation_25e257dc-11ca-4f58-9...	✓ Succeeded
HC_NB_Ingestion_Supplier_be279685-e504-402...	✓ Succeeded
HC_NB_Ingestion_Nation_25e257dc-11ca-4f58-9...	✓ Succeeded
HC_NB_Ingestion_Supplier_be279685-e504-402...	✓ Succeeded
HC_NB_Ingestion_Nation_25e257dc-11ca-4f58-9...	✓ Succeeded
HC_NB_Ingestion_Supplier_be279685-e504-402...	View details ✓ Succeeded
HC_NB_Ingestion_Supplier_be279685-e504-402...	ⓘ ... ✓ Succeeded
HC_NB_Ingestion_Supplier_be279685-e504-402...	✓ Succeeded
HC_NB_Ingestion_Supplier_be279685-e504-402...	✓ Succeeded
HC_NB_Ingestion_Serial_c94d8f0a-1efa-4190-bb...	✓ Succeeded
PL_Orchestration	✓ Succeeded

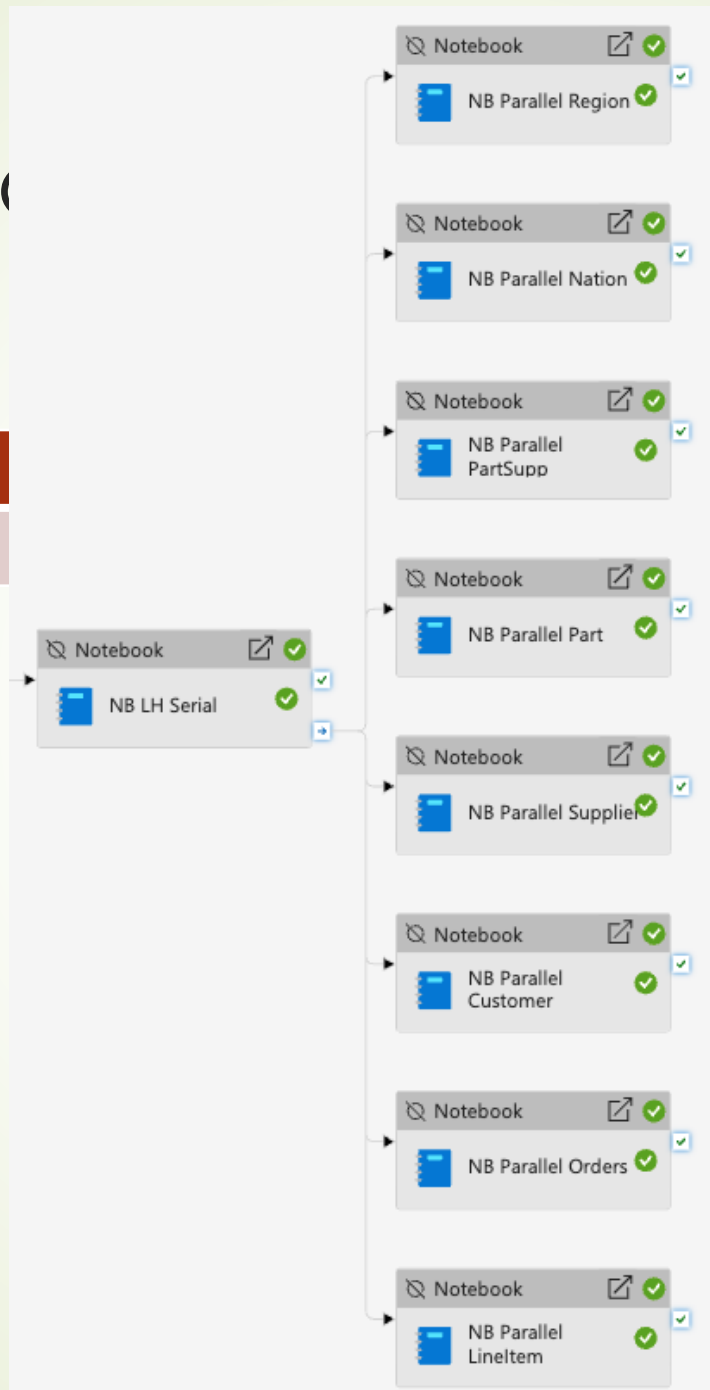
Parallel CU(s)

??

Notebook

Serial Time

24 minutes



Parallel Time

minutes

Parallel CU(s)

??



Why the missing numbers?

High concurrency

It's amalgamated
into the first
notebook

You, yes You!, need
to dive into the
metrics of that
specific notebook
to get to the details

There they are!

PerformanceTesting > NB_Ingestion_Serial F2.0 > HC_NB_Ingestion_Serial F2.0_93169f2d-72f7-459e-9d78-e3ac4270b0d9

Refresh Stop application Monitor run series Spark History Server

Jobs Resources Logs Data **Item snapshots**

Item snapshots <<

Snapshot: NB_Ingestion_Orders F2.0

✓ PL_Orchestration

- > NB_Ingestion_Serial F2.0
- > NB_Ingestion_Region F...
- > NB_Ingestion_Supplier ...
- > NB_Ingestion_Nation N...
- > NB_Ingestion_LineItem ...
- ✓ NB_Ingestion_Orders F...

NB_Ingestion_Orde...

```
1 spark.conf.set("spark.microsoft.delta.optimizeWrite.enabled", "tru
2 spark.conf.set("spark.microsoft.delta.optimizeWrite.binSize", "107
3
4 from pyspark.sql.types import StructType, StructField
5 from pyspark.sql.types import DoubleType, IntegerType, StringType
```

[1] ✓ - Command executed in 560 ms on 1/18/2026, 9:17:46 PM

```
1 LH_SRC_ABFSS = 'abfss://TrialF64@onelake.dfs.fabric.microsoft.com/
Parameters
```

[2] ✓ - Command executed in 434 ms on 1/18/2026, 9:17:47 PM



Differences between Default, Native Execution Engine and Runtime 2.0

File	Default	Native Execution	Fabric 2.0
Serial	22:30	22:37	21:15
region	01:19	00:56	00:45
Nation	01:15	00:56	03:00
PartSupp	08:52	09:30	09:45
Part	13:05	15:16	07:05
Supplier	01:53	01:22	02:10
Customer	04:38	02:31	04:27
Orders	09:25	12:20	10:00
LineItem	18:50	14:11	10:57

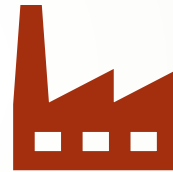
Differences between Default, Native Execution Engine and Runtime 2.0

File	Default	Native Execution	Fabric 2.0
Serial	49.837	85.712	65.250
region	12.542		
Nation	3.062		
PartSupp	13.411		
Part	3.261		21.626
Supplier	40.477	2.123	6.693
Customer	4.839	10.370	6.512
Orders	29.711		13.843
LineItem	12.272		

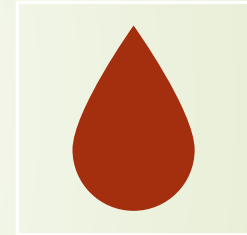
Eventstream to Eventhouse



Limited ingestion because
of max file size that can
be processed



Nation, Region, Supplier



1505 CU(s)

Eventstream to Eventhouse

11,5 GB

768 MB

Get data

Summary

Name

Status

Det

<https://stfabriotpch.blob.core.windows.net/tcphfabric/files/partsupp.tbl.4.csv>



Failed to ingest: Blob size in bytes: '1240521...

Failed to ingest: Blob size in bytes:
'12405216439' has exceeded the size
limit allowed for ingestion ('6442450944'
B)

Tying it all together





Storage cost?

Storage	CU cost	Euro's
Lakehouse	8.758	0,43
Warehouse	10.626	0,53
Eventhouse	4.105	0,20
Fabric SQL DB	64.715	3,23



Ingestion cost

Type	CU cost	Euro's
Pipelines	3.778.380	188,91
Notebooks	169.412	8,47
Event Streams	4.649	2,32
Dataflows Gen2	1.637.230	81,86
Dataflows Gen2 CI/CD	51.560	2,57

NOTE! The results are accurate at the time of measurement. Do not take these as absolute truths but use them as guidance for your own testing!

Numbers shown are averages, outliers will influence the outcomes



Personal storage preferences

1. Lakehouse
2. Eventhouse
3. Warehouse
4. Fabric SQL DB (metadata only)



Is there a best solution



No, but
there is a
best fit

Expertise,
budget,
workforce

It always
depends

Pro Tips



USE FUAM FOR
MONITORING



TEST, TEST AND TEST
AGAIN. NEW RELEASES
CAN GO EITHER WAY



THE CHEAPEST SOLUTION
MAY NOT BE THE BEST
(FOR YOU)

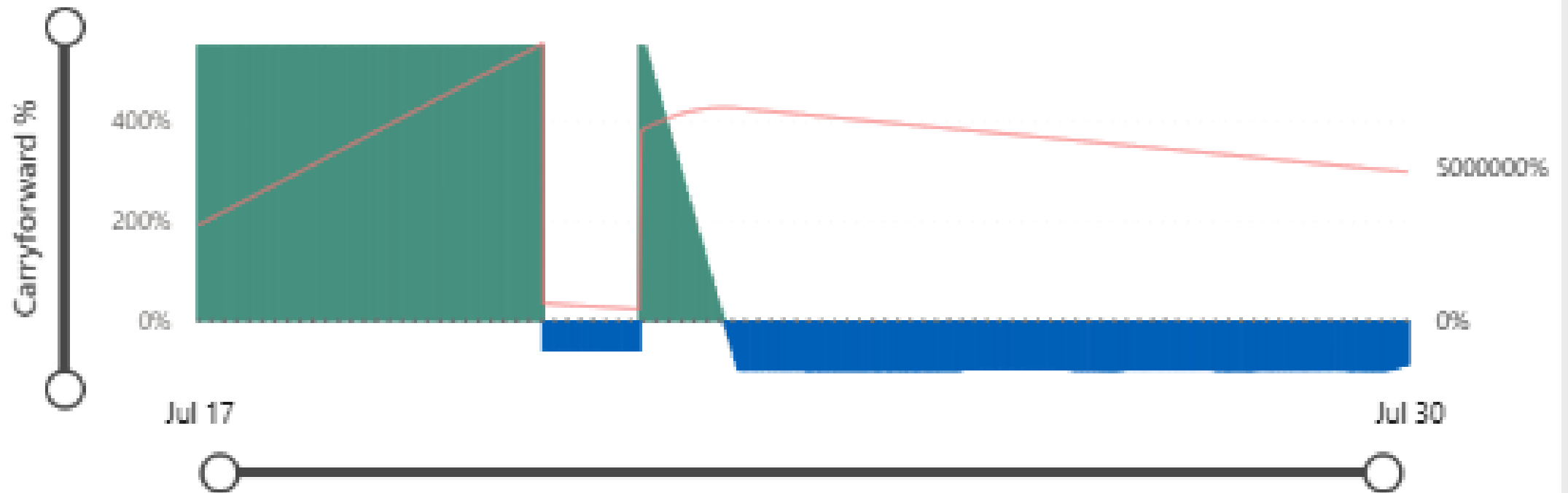
When stuff goes off the rails



➤ 5.000.000% Overage with Always-On in RTI

Overages % over time

● Add % ● Burndown % ● Cumulative (Right axis) %





THANK YOU!